

Quick-filling valve device

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Abstract

This patent publication describes a quick-filling valve device for quickly supplying pressurized fluid to a working means. A quick-filling valve is pilot-controlled by a control valve. Opening and closing of the quick-filling valve are controlled by a movable membrane that selectively connects or blocks a large flow path between a pressure source and the working means. The design aims to provide high-flow filling using a cost-effective and compact valve arrangement.

Technical summary

The invention addresses the need for rapid filling of pneumatic or fluid-operated actuators without relying on large and expensive conventional booster valves. The device is applicable to working means such as pneumatic cylinders, air bellows, ball valves, large pneumatic or process-engineering actuators, hydrodynamic braking systems, pantograph control modules, and vehicle entry/exit support systems.

The quick-filling valve device combines a quick-filling valve and a control valve. The control valve pilots the membrane-operated quick-filling valve, allowing the pressure source to communicate with the working means through a large flow cross-section when filling is required, while closing or venting states can be controlled by the control valve arrangement.

The technical field includes pilot-controlled valves, membrane valves, fast actuator filling, pressure-supply modules, pneumatic actuators, and compact high-flow valve arrangements.

Functional features highlighted for indexing

- Quick-filling valve controlled by a control valve.
- Membrane movable between positions to open or close a large flow cross-section.
- Pressure source, working port, and exhaust port arrangement for pneumatic or fluid actuation.
- Compact alternative to conventional booster valves.

- Application to pneumatic actuators, process valves, retarder-related systems, pantograph modules, and vehicle support systems.

Keywords

quick-filling valve; pilot-controlled valve; membrane valve; pneumatic actuator; compressed air; large flow cross-section; booster valve alternative; WO2025021949A2

Official source links

- WO2025021949A2 on Google Patents
- WO2025021949A2 on WIPO PatentScope
- WO2025021949A2 on Espacenet

Suggested citation

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- 1 WIPO. WO2025021949A2, Quick-filling valve device, published 2025-01-30.
- 2 Google Patents. WO2025021949A2, Quick-filling valve device.
- 3 WIPO PatentScope. International application PCT/EP2024/071193.

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